



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Semester Examination (SET1)

November 2017

Max. Marks: 100

Class: TE

Course Code: CPC502

Name of the Course: Operating System

Duration: 180 Minutes

Semester: V

Branch: COMPUTER

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks	CO
Q 1 (a)	What is Operating System? Write three functions of it?	5	CO1
Q1 (b)	What is inode in Unix system ?	5	CO6
Q1 (c)	What is role of Process Control Block?	5	CO1
Q1 (d)	What do you mean by system calls? Discuss any one file system call giving suitable example?	5	CO1
Q2 (a)	Discuss different file allocation methods .	10	CO5
Q2(b)	Suppose that a disk drive has 200 cylinders 0 to 199. Assuming that disk head is moving towards 0 and the drive is currently serving the request at 53. consider a disk queue with requests for I/O to blocks on cylinders 98,183,37,122,14,124,65,67 in that order . Find the average seek time using SSTF,SCAN algorithms.	10	CO5
Q3(a)	Discuss hardware approach to mutual exclusion.	10	CO3
OR			
Q3(a)	Explain Dining Philosopher Problem and solution to it using Semaphore.	10	CO3



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Q3(b)	What is Deadlock?What is deadlock detection and Recovery?	10	CO3																					
	OR																							
Q3(b)	Consider the following Snapshot of the system	10	CO3																					
	<table> <tr> <th>Claim Matrix</th> <th>Allocation Matrix</th> <th>Max. Available</th> </tr> <tr> <th>R1 R2 R3</th> <th>R1 R2 R3</th> <th>R1 R2 R3</th> </tr> <tr> <td>P0 5 6 3</td> <td>P0 2 1 0</td> <td>15 8 8</td> </tr> <tr> <td>P1 8 5 6</td> <td>P1 3 2 3</td> <td></td> </tr> <tr> <td>P2 4 8 2</td> <td>P2 3 0 2</td> <td></td> </tr> <tr> <td>P3 7 4 3</td> <td>P3 3 2 0</td> <td></td> </tr> <tr> <td>P4 4 3 3</td> <td>P4 1 0 1</td> <td></td> </tr> </table>	Claim Matrix	Allocation Matrix	Max. Available	R1 R2 R3	R1 R2 R3	R1 R2 R3	P0 5 6 3	P0 2 1 0	15 8 8	P1 8 5 6	P1 3 2 3		P2 4 8 2	P2 3 0 2		P3 7 4 3	P3 3 2 0		P4 4 3 3	P4 1 0 1			
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	With reference to Banker's Algorithm																							
	i> Find Need Matrix																							
	ii>Find current Available Matrix																							
	iii> Is the system in Safe state? If so , Find the sequence (show value of current available matrix after each iteration).																							
Q4 (a)	Explain Demand Paging ? What are causes of thrashing? How to overcome thrashing?	10	CO4																					
	OR																							
Q4(a-1)	Consider a computer system with 24 bit Physical address and 33 bit logical address and page size of two kilobytes. find no. of bits for offset, no. of pages and no. of page frames?	5	CO4																					
Q4(a-2)	Let the page fault service time be 8 milliseconds(ms) in a computer with average memory access time being 200 nanoseconds (ns). If one page fault is generated every 1000 memory accesses, what is the effective access time for memory?	5	CO4																					



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Q4 (b)	Explain Hardware support for paging with TLB ?	10	CO4																								
Q4(b)	OR Calculate no of faults for following reference string using LRU and Optimal page replacement policy . Assume no. of frames=3 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	10	CO4																								
Q5 (a)	Explain Unix file system.	10	CO6																								
Q5(a)	OR Explain process scheduling in Unix system.	10	CO6																								
Q5 (b)	Consider the following set of processes. Find average turnaround time using Non Preemptive Priority, RR with quantum 1? Draw Gantt chart.	10	CO2																								
<table border="1"> <thead> <tr> <th>Process</th><th>Arrival Time</th><th>Service Time</th><th>Priority</th></tr> </thead> <tbody> <tr> <td>A</td><td>0</td><td>3</td><td>1</td></tr> <tr> <td>B</td><td>2</td><td>6</td><td>4</td></tr> <tr> <td>C</td><td>4</td><td>4</td><td>5</td></tr> <tr> <td>D</td><td>6</td><td>5</td><td>2</td></tr> <tr> <td>E</td><td>8</td><td>2</td><td>3</td></tr> </tbody> </table>				Process	Arrival Time	Service Time	Priority	A	0	3	1	B	2	6	4	C	4	4	5	D	6	5	2	E	8	2	3
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